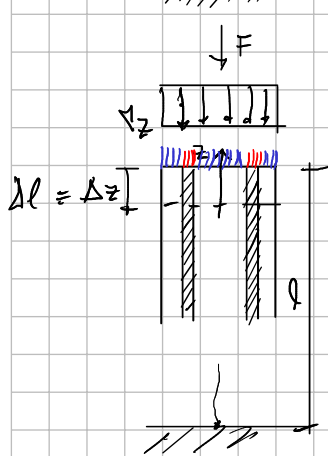


$$\frac{E_a}{E_h} = 1.5$$



$$\begin{cases} \Delta l_a = \Delta l_h \\ l_a = l_h \end{cases}$$

$$\epsilon_{zz} = \frac{\partial u_z}{\partial z} \rightarrow du_z = \epsilon_{zz} dz$$

$$\Delta l = u_z(l) - u_z(0) = \epsilon_{zz} \cdot \frac{l}{\epsilon_{zz}} \rightarrow \epsilon_{zz} = \frac{\Delta l}{l}$$

$$\epsilon_{zza} = \epsilon_{zzh} = \frac{\Delta l_a}{l_a} = \frac{\Delta l_h}{l_h} = \frac{\Delta l}{l}$$



ley de Hooke:

$$\epsilon_{zz} = \frac{1}{E} [\sigma_{zz} - \nu (\sigma_{xx} + \sigma_{yy})]$$

→ no tiene restricción lateral

$$\epsilon_{zz} = \frac{\sigma_{zz}}{E}$$

$$\epsilon_{zza} = \frac{\sigma_{zza}}{E_a}$$

$$\sigma_{zza} = \frac{F_a}{A_a}$$

$$\epsilon_{zzh} = \frac{\sigma_{zzh}}{E_h}$$

$$\sigma_{zzh} = \frac{F_h}{A_h}$$

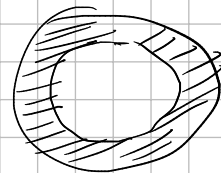
$$\epsilon_{zza} = \frac{F_a}{A_a E_a}$$

$$\epsilon_{zzh} = \frac{F_h}{A_h E_h}$$

$$\epsilon_{zza} = \epsilon_{zzh}$$

$$\frac{F_a}{F_h} = \frac{A_a E_a}{A_h E_h}$$

sección Anular



$$A_a = 36 \times 6.5 \text{ cm}^2$$

$$A_h = \pi (r_e^2 - r_i^2) - 4a$$

$$= 2298 \text{ cm}^2$$

$$\frac{F_a}{F_h} = 1.53$$

$$F = F_h (1 + 1.53)$$

$$r = \|\underline{r}\|$$

$$r = \sqrt{x_j \cdot x_j}$$

$$\frac{\partial r}{\partial x_i}$$

Martes

→ 17hs

